

E1: Analytic Identifiability Engine

How it works and how it was validated

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Amir Dollie (DLLAMI004), EEE4022S

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Summary

What E1 is. For any floe, sensor layout, sea state, sensor noise and record length, E1 predicts how accurately the inverse can recover floe stiffness β and radius R , the size of systematic bias, and whether the floe is identifiable, without simulating time records. **Outcome: 16/16 checks passed.** Predicted uncertainties matched trusted Monte Carlo/Fisher targets within 0.8–4.0% (5% tolerance) and Welch-bias distance within 6–7% (10% tolerance). **Why it matters.** The remaining programme needs hundreds to thousands of evaluations: a new floe costs about 2.5 min, while changes in layout, noise, record length or sea spectrum are then mostly algebra. **Caveat.** E1 is a local Fisher prediction; in G3 the realised nonlinear inversion scatter was 4–11% larger, so E1 uncertainties are slightly optimistic.

1 Purpose and reference setting

Gate G3 showed that the frozen inverse can recover $\theta = [\ln \beta, \ln R]^T$ from multi-IMU transmissibilities, with draught parameter γ known and fixed. G3, however, required 200–400 synthetic records per parameter point. The remaining studies ask which layout is best (D1), which floes are identifiable (A1), how noise (R1), sea state (R2) and record duration (D2) matter, and how wrong noise-floor or draught assumptions propagate (R3, R4). Monte Carlo at every point is not feasible, so E1 replaces record-by-record simulation with the analytic map

$$\{p, \text{layout}, S_u, S_{nn}, N, L\} \longrightarrow \{\sigma_{\ln \beta}, \sigma_{\ln R}, \rho, \kappa(F), d_W, d_{\text{sys}}, d_{\text{tot}}, \text{coherence}, \text{class}\}.$$

It uses the same EMM and transmissibility estimator as G3 and is allowed to support the studies only after reproducing the trusted results.

Reference item	Value	Reference item	Value
Scaling	$H = 1.88 \text{ m}; \beta = D/(\rho_w g H^4), \gamma = d/H, R = R_{\text{phys}}/H$	Record	$N = 65536, \Delta t = 0.1 \text{ s (109 min)}$
Nominal p_0	$\beta_0 = 4.70 \times 10^{-5}, \gamma_0 = 1.45 \times 10^{-3}, R_0 = 0.383 (R_{\text{phys}} = 0.72 \text{ m})$	Welch	Hann, $L = 2048$, 50% overlap, 63 segments
Bins	every second bin; 84 bins in 3.178–8.301 rad/s	Noise	G3 $S_{nn} = 1.416 \times 10^{-6}$
Layout	$(0.3R, 0^\circ), (0.3R, 180^\circ), (0.5R, 180^\circ), (0.9R, 0^\circ)$	Reference	outermost sensor at $0.9R$

2 Engine calculation

1. EMM field and parameter sensitivities — `emmField.m`. The EMM returns acceleration FRFs $H_j(\omega; p)$ at fixed sensor locations. It is solved on 29 nodes spaced 0.2 rad/s apart and spline-interpolated to the retained bins. Central differences in log-parameters give

$$\frac{\partial H_j}{\partial \ln p_i} \approx \frac{H_j(p_i e^{+h}) - H_j(p_i e^{-h})}{2h}.$$

Sensor positions remain fixed physically when R is perturbed, so the derivative measures a change in the floe, not a change in layout. For the frozen inverse, one scenario requires nominal, $\beta^\pm$ and $R^\pm$: five field evaluations, about 30 s each in MATLAB. The same field may contain all 33 D1 candidate positions.

2. Transmissibilities and Jacobian — `transmissibilityFromField.m`. The inverse measures $T_j = H_j/H_r$, with sensitivity

$$\frac{\partial T_j}{\partial \ln p_i} = \frac{(\partial H_j / \partial \ln p_i) H_r - H_j (\partial H_r / \partial \ln p_i)}{H_r^2}.$$

At each frequency, the real and imaginary parts of the three non-reference transmissibilities form a six-element feature vector f_k , and $J_k = \partial f_k / \partial \theta$ is 6×2 .

3. Finite-record transmissibility covariance — `analyticTransmissibilityCovariance.m`. The corrected estimator is $\hat{T}_j = \hat{S}_{jr}/D$, $D = \hat{S}_{rr} - S_{nn,r}$. Treating the Welch spectral matrix as complex Wishart and linearising the ratio gives

$$C_{jl} = \frac{S_{rr}(S_{jl} - \bar{T}_l S_{jr} - T_j S_{rl} + T_j \bar{T}_l S_{rr})}{nD^2}, \quad P_{jl} = \frac{(S_{jr} - T_j S_{rr})(S_{lr} - \bar{T}_l S_{rr})}{nD^2}.$$

The real covariance block Σ_k for $[\Re T_1, \Re T_2, \Re T_3, \Im T_1, \Im T_2, \Im T_3]^T$ is

$$\Sigma_{\Re\Re} = \frac{1}{2}\Re(C + P), \quad \Sigma_{\Im\Im} = \frac{1}{2}\Re(C - P), \quad \Sigma_{\Re\Im} = \frac{1}{2}\Im(P - C).$$

Thus E1 retains shared-reference correlations between T_j and Re/Im correlations; independent scalar variances would make layout comparisons too optimistic.

Two refinements were required. `welchSmoothedSpectrum.m` uses the Hann-window-smoothed spectral matrix rather than the bin-centre value because $T(\omega)$ varies across the window; without this, σ was 12% too optimistic at $L = 512$ and d_W was wrong. `welchNEff.m` gives the overlap-corrected $N_{\text{eff}} = 59.734$; E1 uses $n = N_{\text{eff}} - 1$, treated as an empirically validated finite-sample correction rather than a derived exact result.

4. Fisher uncertainty — `fisherFromBlocks.m`. Only bins accepted by the estimator contribute:

$$F = \sum_{k \in \text{accepted}} J_k^T \Sigma_k^{-1} J_k, \quad C_\theta = F^{-1}, \quad \sigma_{\ln \beta} = \sqrt{(F^{-1})_{11}}, \quad \sigma_{\ln R} = \sqrt{(F^{-1})_{22}}.$$

Because the parameters are logarithms, $\sigma_{\ln \beta} = 0.005$ means about 0.5% uncertainty in β . C_θ also gives the correlation ρ , while $\kappa(F)$ measures parameter degeneracy.

5. Bias projection — `welchExpectedTransmissibility.m`. A feature bias b produces

$$\delta\theta = F^{-1} J^T \Sigma^{-1} b, \quad d = \sqrt{\delta\theta^T F \delta\theta}.$$

$d < 0.5$ means a shift small relative to random scatter; $d > 1$ means it dominates. E1 includes Welch-resolution bias b_W and deliberate systematic errors b_{sys} , e.g. a wrong draught in R4 with $b_{\text{sys}} \approx -J_\gamma \ln(1 + \epsilon_\gamma)$. Bias vectors add before significance is evaluated: $\delta\theta_{\text{tot}} = \delta\theta_W + \delta\theta_{\text{sys}}$.

6. Classification — `evaluateScenario.m`.

Class	$s_{\text{max}} = \max(\sigma_{\ln \beta}, \sigma_{\ln R})$	d_{tot}	$\kappa(F)$
Identifiable	$\leq 1\%$	≤ 0.5	$\leq 10^8$
Marginal	$\leq 5\%$	≤ 1	$\leq 10^8$
Not identifiable	anything else, or singular F		

Safeguards built into the engine

Rejected bins: if $D < 0.1S_{rr}$, the real estimator rejects the bin, so E1 removes it from F and every bias projection; otherwise high-noise R1 cases would look too accurate. **Singular F :** no regularisation is used; a floe with no information about β returns $\sigma = \infty$ and *Not identifiable*, making failure a valid result rather than a crash. **γ is never estimated:** the estimated vector is locked to $[\ln \beta, \ln R]$ in that order. J_γ may be computed only as a separate nuisance sensitivity for R4.

3 Validation gate: 16 of 16 passed

Targets came from the U1 200-record covariance Monte Carlo, the $L = 2048$ bias check and gate G3. The production-chain gate was run in MATLAB from an empty cache, so the EMM fields at p_0 , A and B were recomputed.

Check	What it shows	Result
Covariance formula vs 20,000 simulated Wishart draws	C , P and Σ algebra is correct with noisy reference	variance ratios 0.994–1.016; Frobenius error 1.2% PASS
Σ_k vs 200-record MC, every element over 84 bins	full variance and shared-reference correlation structure is reproduced within MC sampling noise	z rms 1.02 (variances), 1.02 (covariances); pure noise: 1 PASS
Effective segments	analytic and estimator DOF agree	$N_{\text{eff}} = 59.734$ (63 segments) PASS
Non-reference sensor permutation	result is independent of sensor labels	relative change in F : 1.9×10^{-16} PASS
DOF scaling	information grows with record information as expected	$F \times 2.017$ PASS
γ nuisance / estimation guard	J_γ can be computed without entering F ; attempts to estimate or reorder parameters are refused	F remains 2×2 ; invalid requests refused PASS
Rejected-bin test	only estimator-accepted bins enter information	78/84 used; F matches hand-built valid-bin result PASS
No- β field	true non-identifiability is handled cleanly	$\sigma = [\infty, \infty]$, Not identifiable PASS

Parameter-uncertainty gate (5% tolerance).

Point	$\sigma_{\ln \beta}$			$\sigma_{\ln R}$		
	E1	Target	Diff.	E1	Target	Diff.
$p_0, L = 2048$	.00468	.00462	+1.3%	.000889	.000880	+1.0%
$p_0, L = 1024$	.00479	.00470	+1.9%	.000896	.000880	+1.8%
$p_0, L = 512$	.00370	.00366	+1.2%	.000665	.000660	+0.8%
A: $1.2\beta_0, 0.97R_0$	.00502	.00498	+0.8%	.000962	.000950	+1.3%
B: $0.8\beta_0, 1.03R_0$	.00445	.00431	+3.1%	.000852	.000820	+4.0%

At the three production points ($L = 2048$), $(\rho, \kappa(F), \text{class})$ is (0.842, 101, Identifiable) at p_0 , (0.874, 122, Identifiable) at A, and (0.817, 86, Identifiable) at B.

Welch-bias gate (10% tolerance). $L = 2048$: $d_W = 0.244$ vs 0.26 (−6%); $L = 1024$: 0.842 vs 0.91 (−7%); $L = 512$: 3.41 vs 3.68 (−7%). The steep fall with window length explains the production choice $L = 2048$: there the bias is about one quarter of a random-uncertainty unit, while at $L = 512$ it dominates.

4 Interpretation, limitations and use

- What was matched.** The σ targets are Fisher predictions using a covariance estimated from 200 Monte Carlo records. E1 is consistently 1–4% higher. Inverting a covariance estimated from 200 records with six features overstates F by roughly 200/193, so the MC-based σ is about 2% too small.
- Agreement is not perfectly uniform.** Error rises from about 1% at p_0 to 4% at B, the closest result to the 5% gate.
- E1 is local.** It reproduces the Fisher prediction, not realised nonlinear inversion scatter. G3 scatter was 1.04–1.11 times the Fisher prediction, so E1 is probably 5–10% optimistic for the full estimator. M1 checks selected extreme scenarios by Monte Carlo.
- Bias tolerance amendment.** The preregistration fixed the d_W targets but not a tolerance. The 10% tolerance was added after the first comparison and logged in `cfg.meta.changes`. All three differences have the same sign (−6 to −7%), indicating a small consistent under-prediction.
- Validation scope.** Direct gating covers basin depth, Level 2C, G3 noise and U1 sea spectrum over +20/−20% in β and −3/+3% in R . Farther floes, deep water and severe noise use the same derivation but are not directly gated; M1 provides selected checks.

The engine is now frozen unless a later study exposes a defect. Each remaining study becomes a sweep of `evaluateScenario`:

Study	Question	How E1 makes it feasible
D1	best sensor layout/reference?	field once on 33 points; thousands of layouts are algebra
A1	which floes are identifiable and where does failure occur?	24 floes at about 2.5 min each; failure is a valid result
R1 / R2 / D2	tolerable noise / effect of sea state / required duration?	noise, spectrum and record-length changes need no new EMM field
R3 / R4	wrong noise floor or draught?	project deliberate feature bias; use nuisance J_γ for R4
M1	does the analytic engine remain valid at extremes?	Monte Carlo only at a few selected scenarios

What E1 enables

Before E1 the project could show that the inverse works at a small number of parameter points. E1 makes it possible to determine *where* the inverse works, *how accurately*, *what controls that accuracy*, and *where it stops working*. The 16/16 validation gate is what justifies using the engine for those conclusions.